

		5
30	B	For X: $\left(\frac{1}{2}\right)^{\frac{t}{T_x}} = \frac{1}{8} \Rightarrow \frac{t}{T_x} = \frac{\lg(1/8)}{\lg(1/2)} = 3 \Rightarrow T_x = \frac{t}{3}$ For Y: $\left(\frac{1}{2}\right)^{\frac{t}{T_y}} = \frac{1}{4} \Rightarrow \frac{t}{T_y} = \frac{\lg(1/4)}{\lg(1/2)} = 2 \Rightarrow T_y = \frac{t}{2}$ $\frac{T_x}{T_y} = \frac{2}{3} = 0.66667 = 0.67$